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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Bubble Sort
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data sequentially
- c. Rearranging data without sorting
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. order()
b. arrange()
c. sort()
d. sorted()

The correct answer is: sort()

Question **4**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It makes the data unsorted
c. It slows down the process
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **5**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching
b. Sorting
c. Complexity
d. Merging

The correct answer is:
Complexity

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting decreases the efficiency of selection operations
- c. Sorting makes it harder to search for items
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **7**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort does not guarantee sorted order
- b. Bubble Sort is difficult to implement
- c. Bubble Sort cannot handle duplicate elements
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **8**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Sorting data in a non-sequential manner
- c. Using the divide-and-conquer approach
- d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Finding a specific element in a list
- c. Merging two datasets
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. order()
- c. sorted()
- d. sort()

The correct answer is: sorted()

Question **11**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It has no effect on the analysis process
- b. It makes the analysis slower
- c. It complicates the analysis process
- d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **12**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction

b. Insertion

c. Selection

d. Distribution

The correct answer is:
Insertion

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data

b. It simplifies the sorting process by using only one pass

c. It only works on small datasets

d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others

b. Combining sorted sublists

c. Dividing the input into smaller subproblems

d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **15**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach
b. It is less efficient than bubble sort
c. It does not require recursion
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach